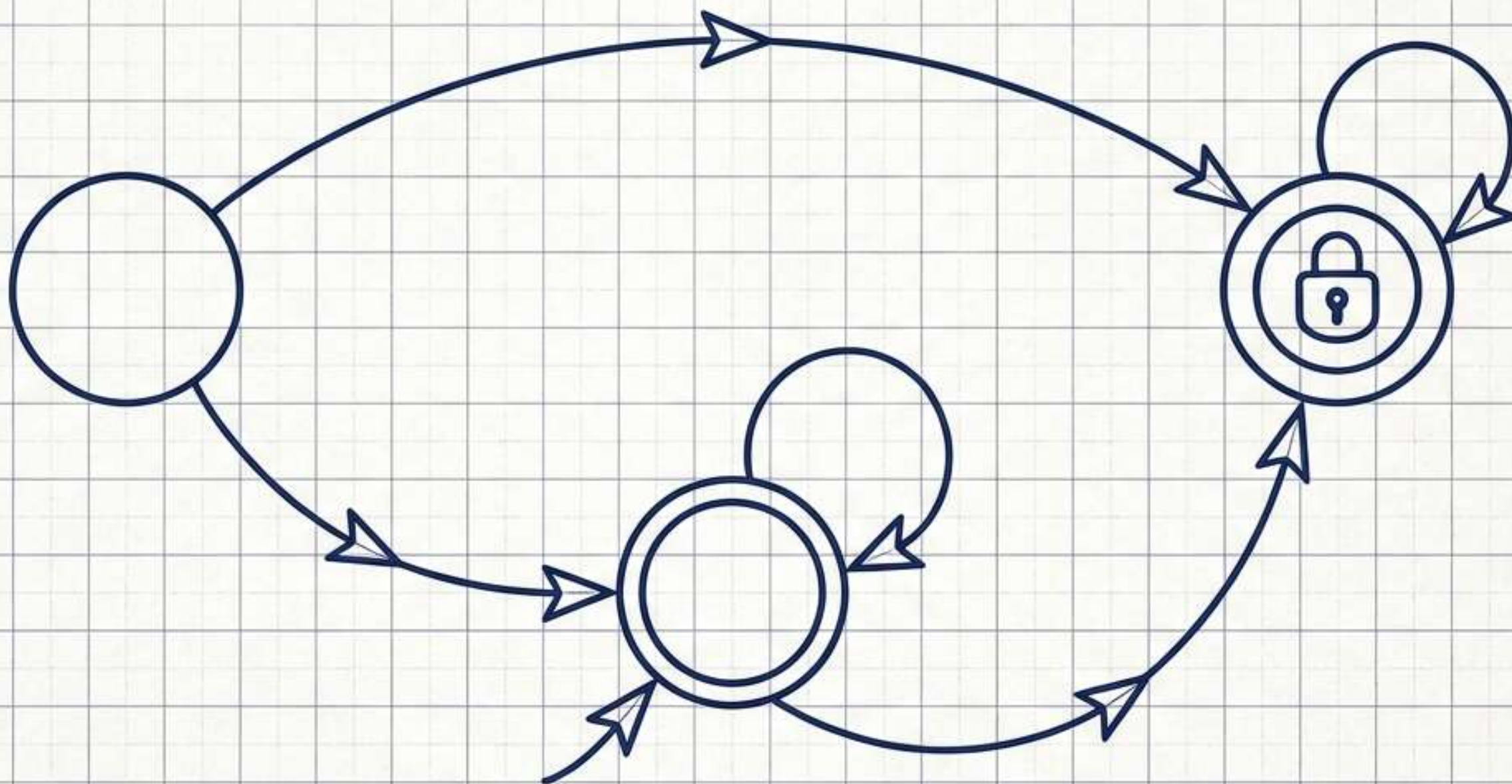


DESIGNING A DETERMINISTIC FINITE AUTOMATON

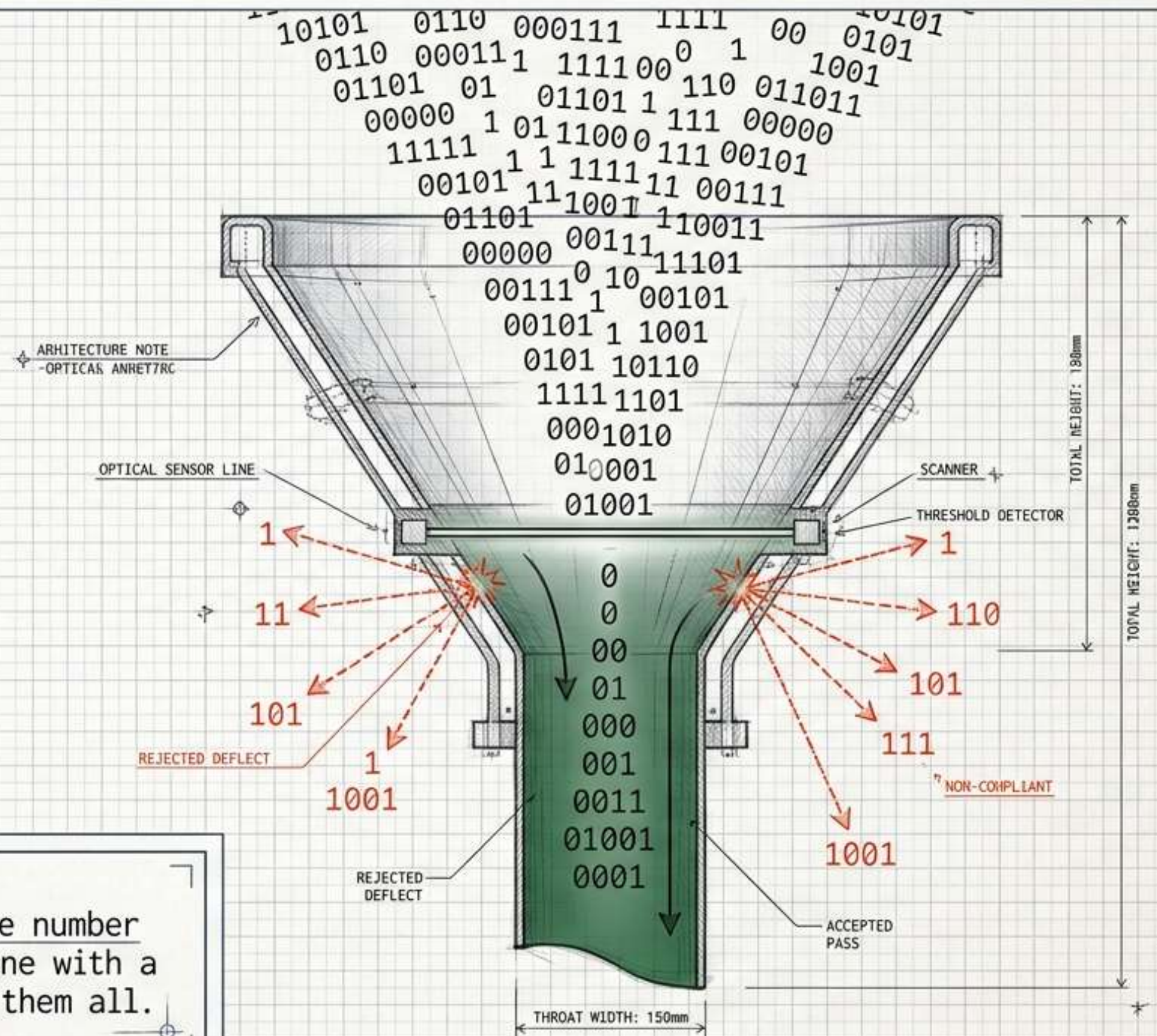
A step-by-step visual guide to building a state machine for strings starting with "0".



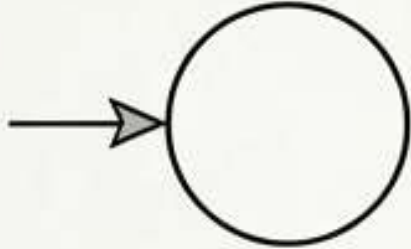
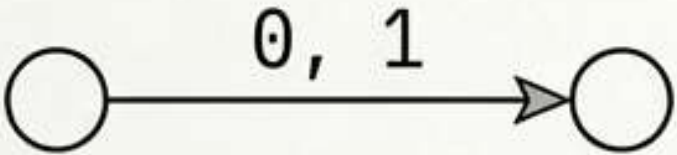
Language L1 Definition:

The set of all
strings that
start with 0.

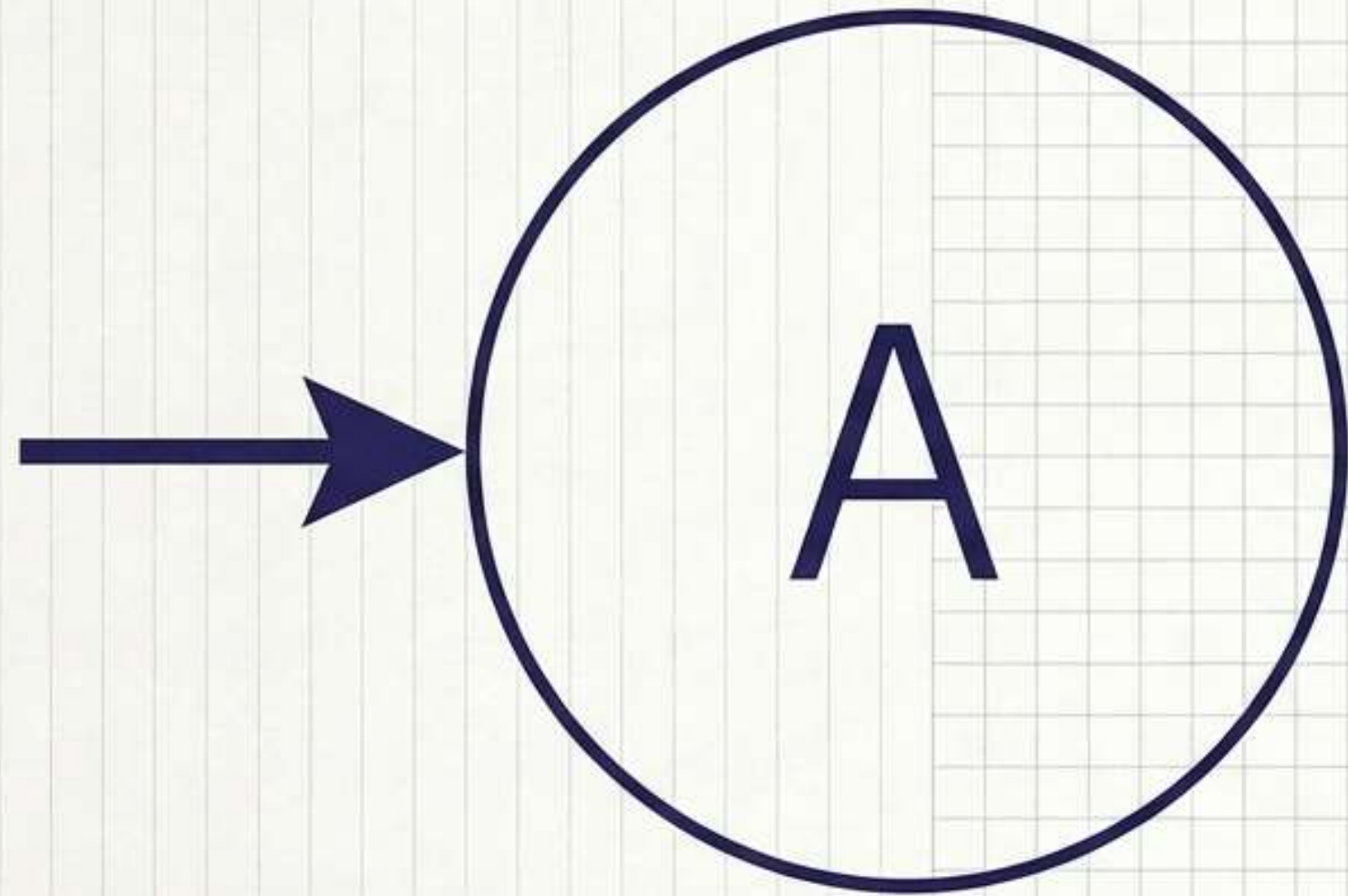
Because this language contains an infinite number
of possible strings, we must build a machine with a
finite number of rules to accurately sort them all.



The Visual Vocabulary of a State Machine

Schematic Icon	Component Name	Architectural Function
	Initial (Start) State	The starting point for every incoming string.
	Final (Accept) State	If a string ends its journey here, it belongs in the language.
	Transition Pathway	The exact route taken when a specific binary input is read.

The Golden Rule: In a complete DFA, every single state **MUST** have exactly one transition pathway for every possible input (0 and 1).



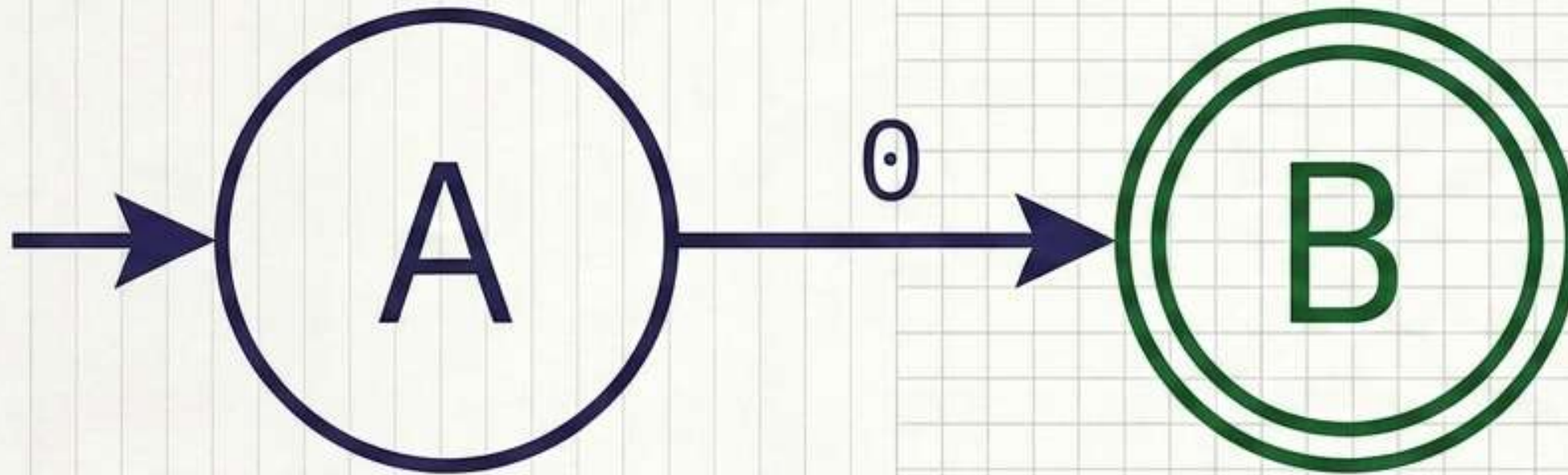
Current State:
A (Initial State)

Logic:

Every string begins its journey exactly here. At this moment, the machine is suspended, waiting to read the very first digit of the incoming string.

Pending Decisions:

Needs a path for input '0', and a path for input '1'.

**Trigger:**

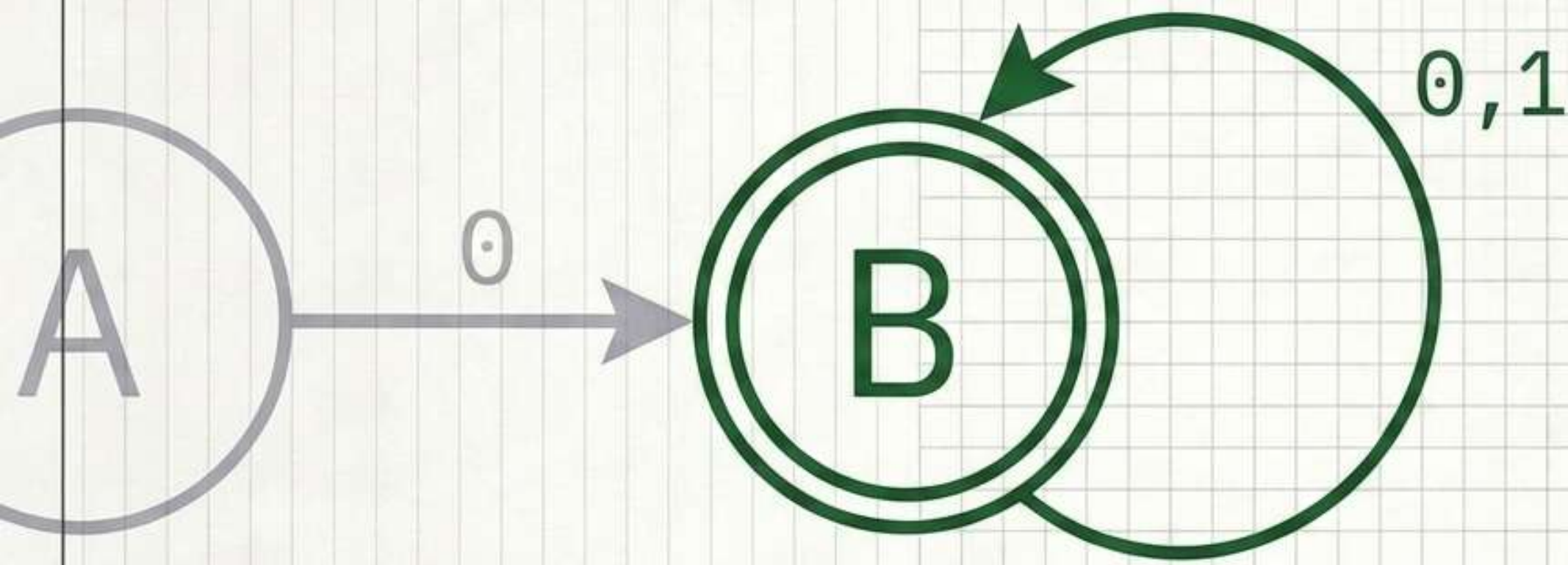
Transition on '0'

Logic:

If the first digit of the input is '0', the core condition of Language L1 is instantly satisfied.

Action:

The machine routes the string into State B, the Final (Accepting) State. The string is currently valid.



Sustaining Acceptance

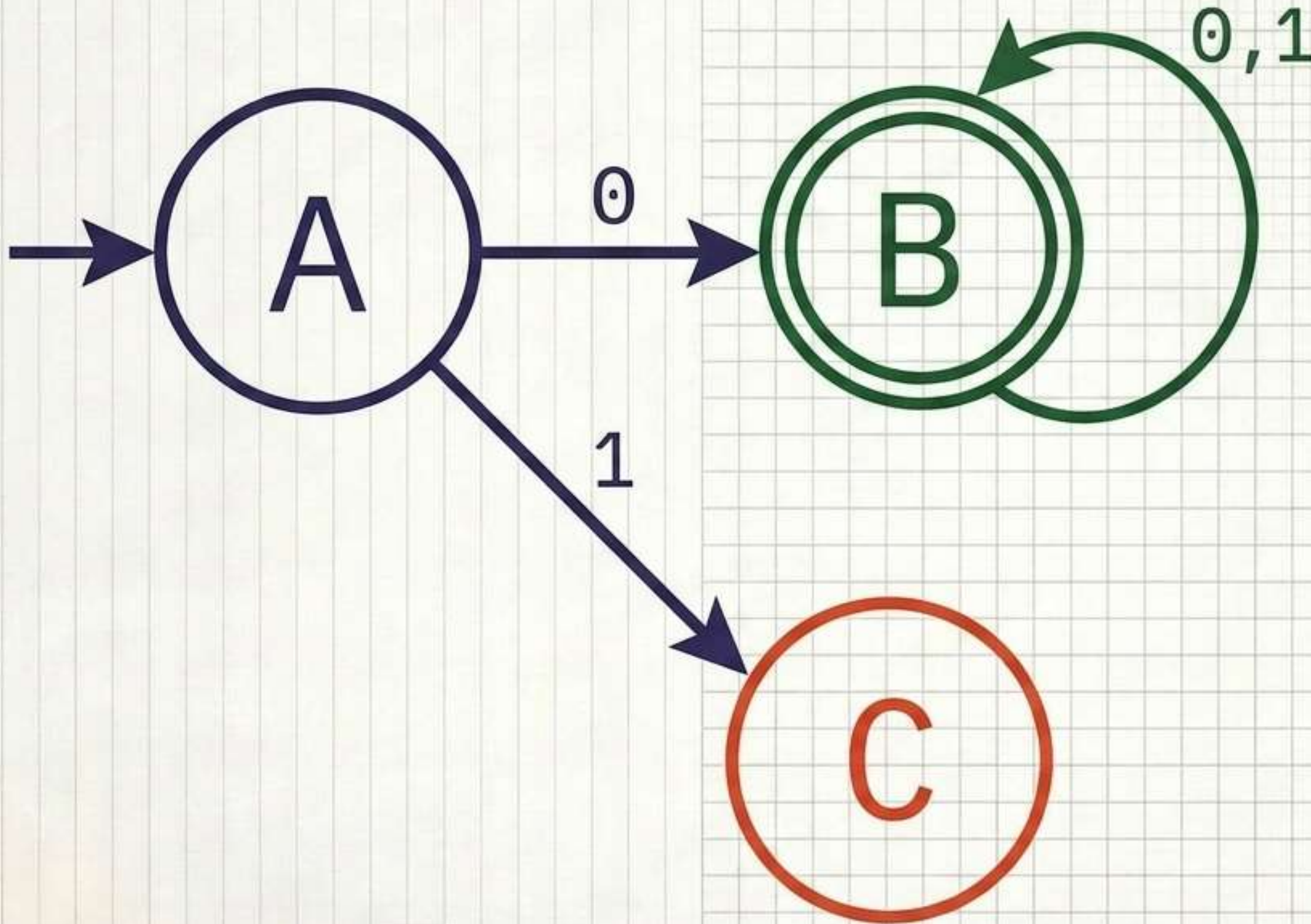
Trigger: Inputs '0' or '1' while in State B

Logic:

Our only rule is 'starts with 0'. Therefore, any characters that follow—whether zeroes or ones—are completely irrelevant to the string's validity.

Action:

Once a string enters State B, it triggers a self-loop. It will stay in State B forever, ensuring the string remains accepted no matter its length.



Handling an Invalid Start

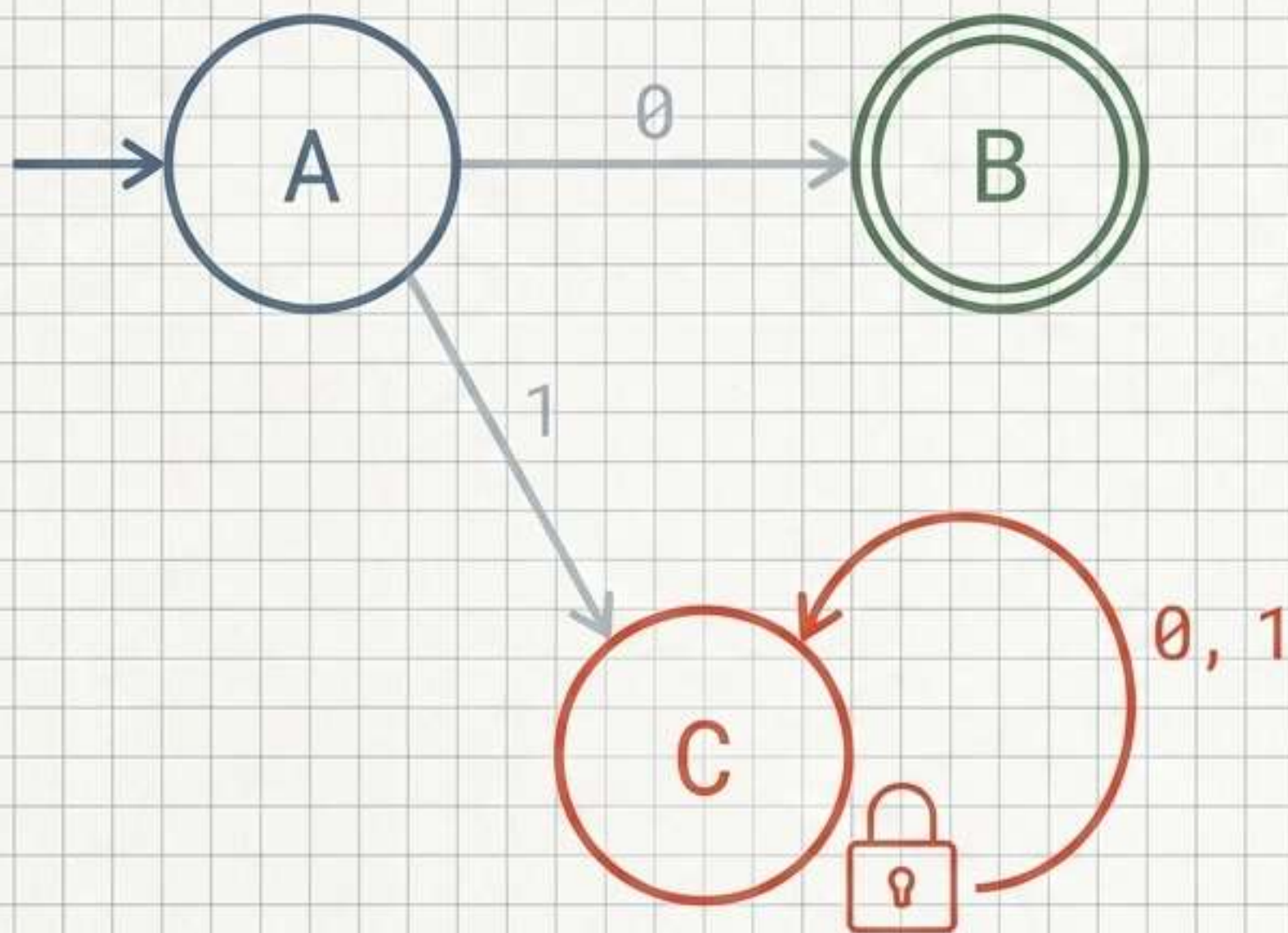
Trigger: Transition on '1' from Initial State

Logic:

If the very first digit is '1', the core rule of Language L1 is immediately and permanently broken.

Action:

The machine must divert this string away from the Accept state. It is forcefully routed downwards into State C.



Inspector Panel

Current State:

C (The Trap / Dead State)

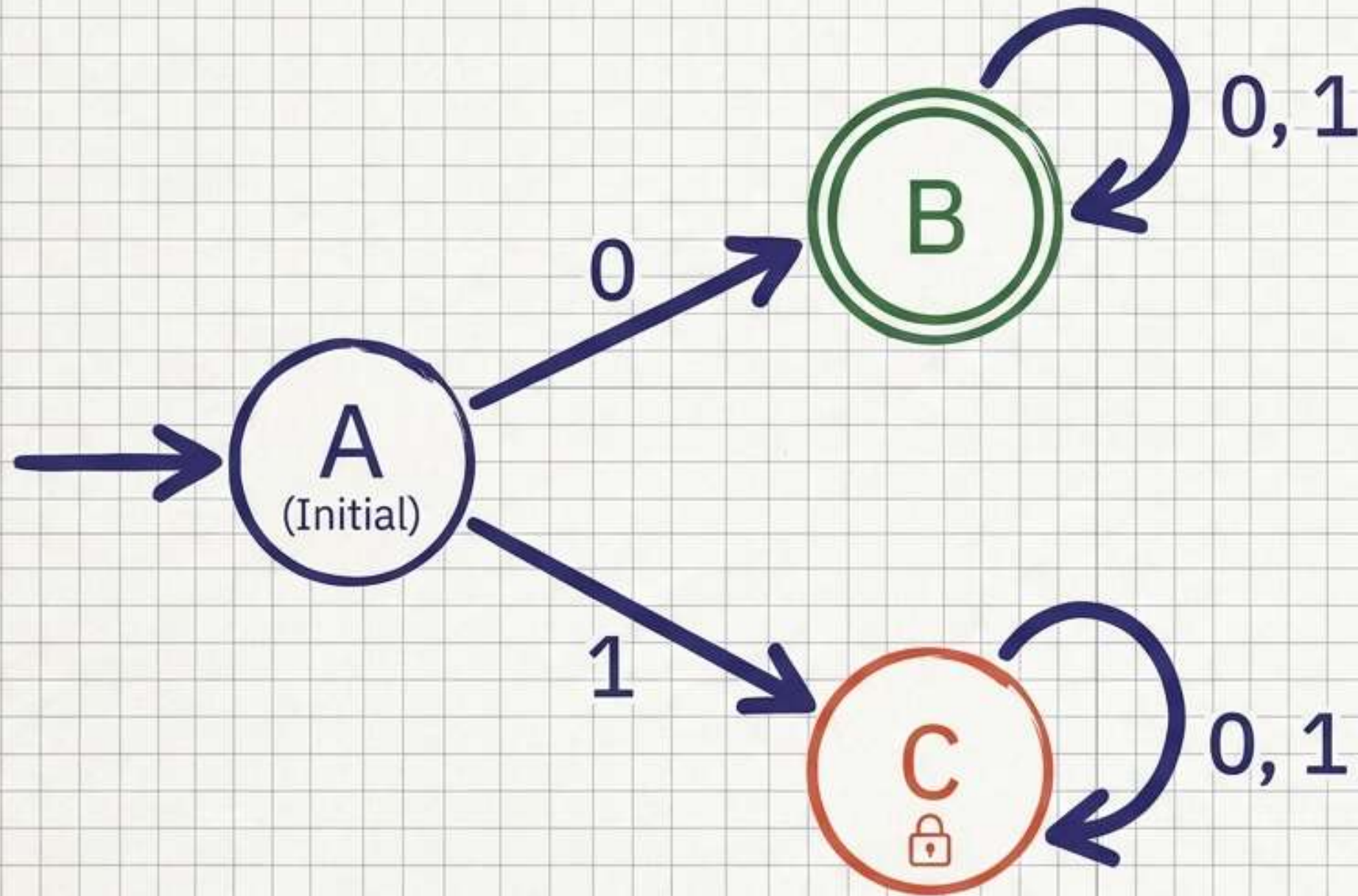
Logic:

Once a string starts with a '1', no combination of subsequent zeroes or ones can magically fix it. It is permanently invalid.

Action:

Any input received in State C triggers a self-loop. The string is trapped outside the Final State and will ultimately be rejected.

The Complete Architectural Blueprint



Inspector Panel

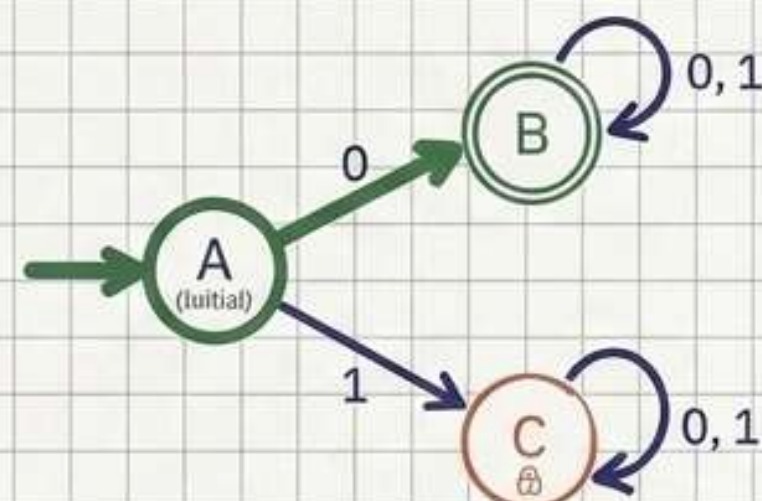
System Status: Mathematically Complete.

Verification: Every single state (A, B, and C) now possesses exactly one outgoing pathway for every possible input in the alphabet $\{0, 1\}$.

Result: Zero ambiguity. The machine can process any binary string of any length.

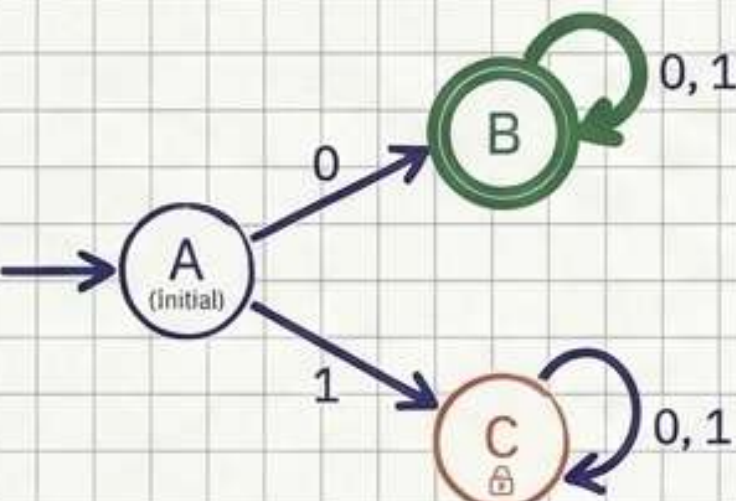
Trace Verification: Valid Input '001'

Input: **[0]** 0 1



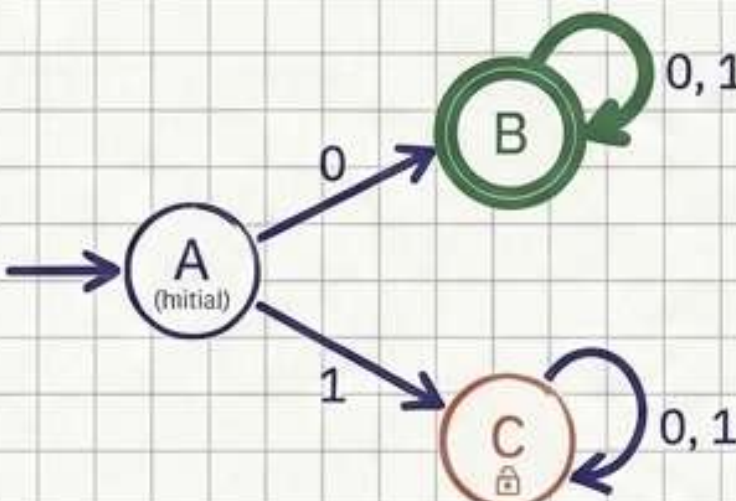
Current: A -> Next: B

Input: 0 **[0]** 1



Current: B -> Next: B

Input: 0 0 **[1]**



Current: B -> Next: B

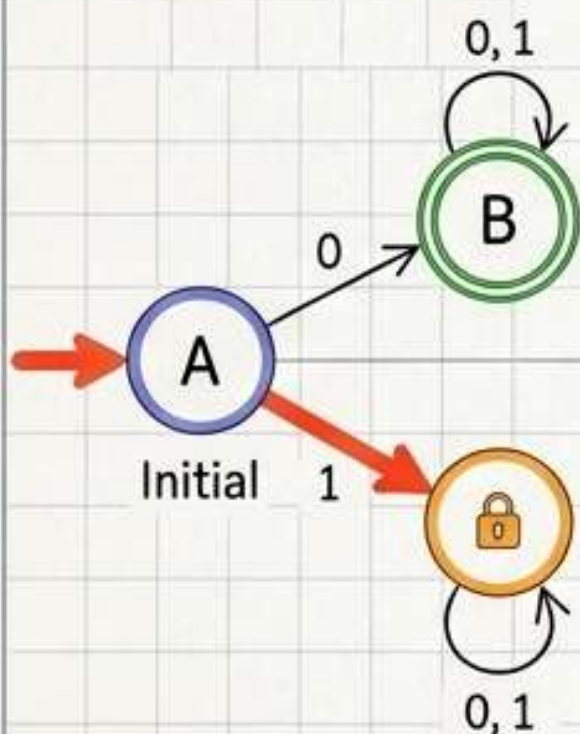
End of string.



Status: **ACCEPTED**

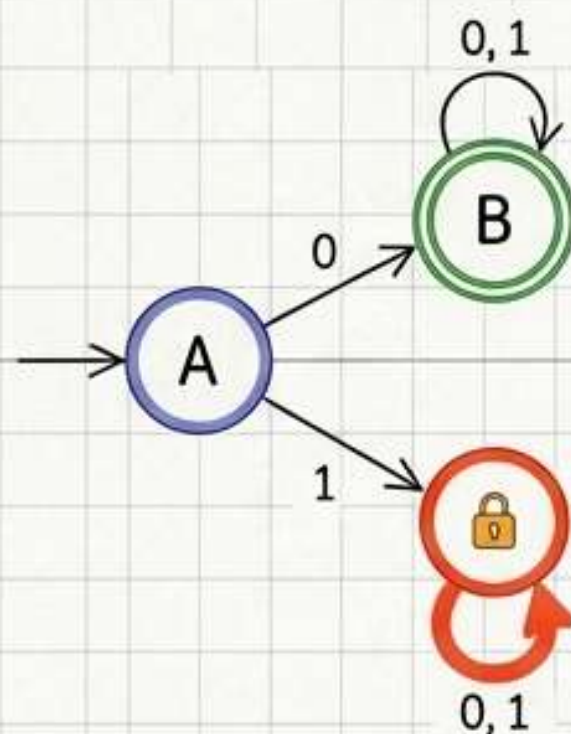
The string properly started with 0 and terminated in a final state.

Input: [1] 0 1



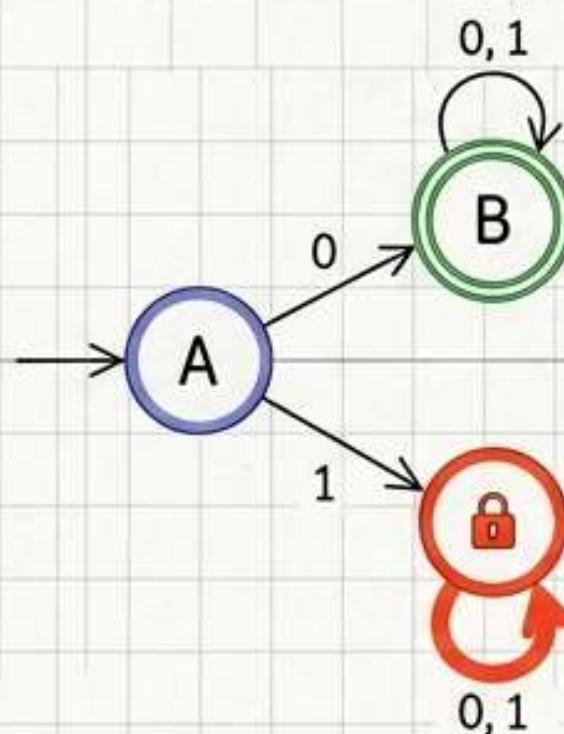
Current: A -> Next: C

Input: 1 [0] 1



Current: C -> Next: C

Input: 1 0 [1]



Current: C -> Next: C

End of string.



Status: REJECTED

The string was
successfully trapped
outside the final state.

Endpoint Diagnostic: Final vs. Trap

State B



Role: Final / Accept State

Visual Indicator: Double Circle

Entry Condition: String's first digit is 0

Exit Condition: None (Permanent self-loop)

System Result: Acceptance

State C



Role: Trap / Dead State

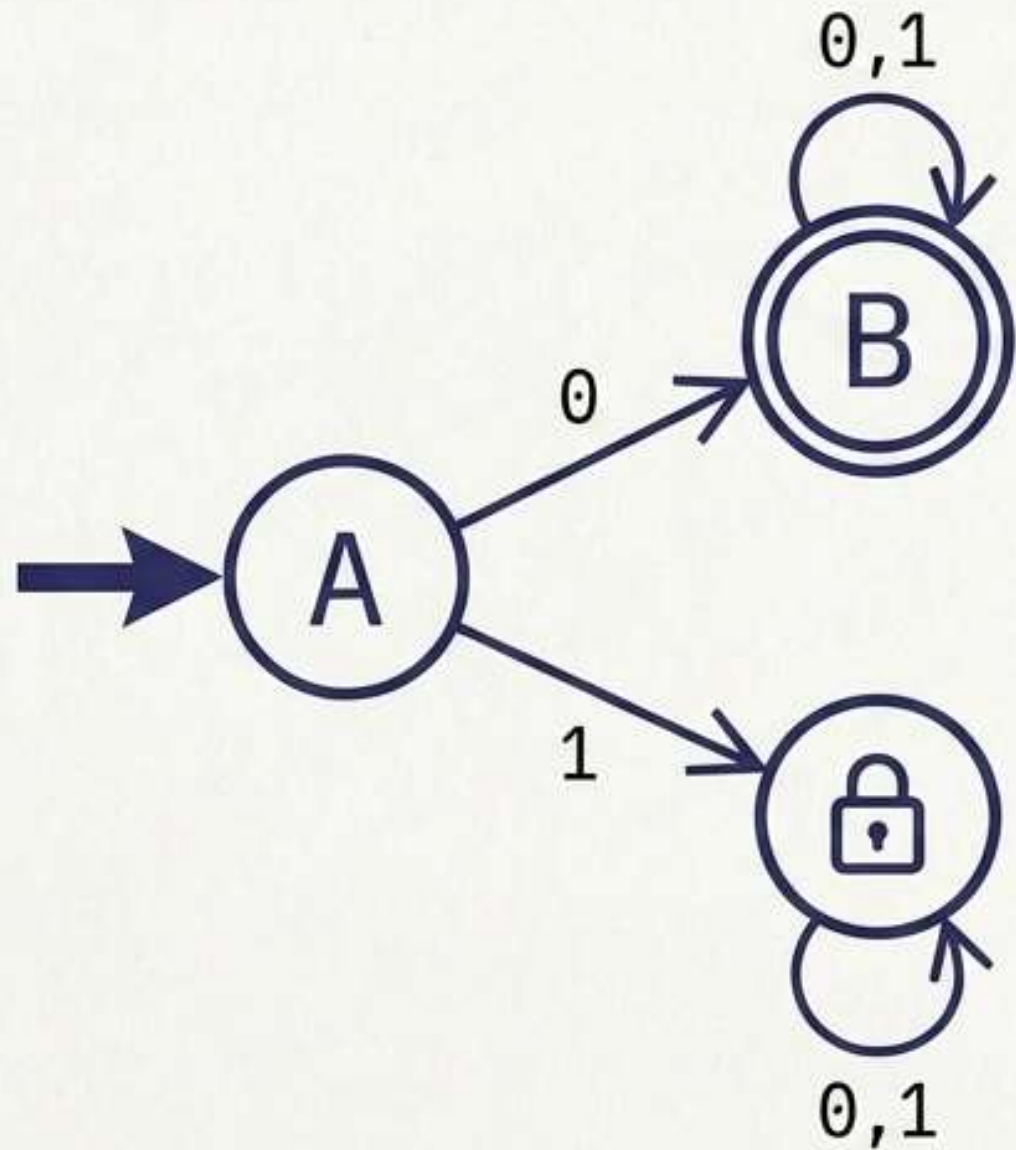
Visual Indicator: Single Circle (with Padlock)

Entry Condition: String's first digit is 1

Exit Condition: None (Permanent self-loop)

System Result: Rejection

Distilling Geometry into Logic



State	Input 0	Input 1
-> A	B	C
* B	B	B
C	C	C

-> denotes start state
* denotes final state

Every visual finite automaton translates perfectly into a programmable routing table. We have successfully engineered a theoretical mathematical constraint into a functional system.